

Applied Quantum Algorithms

Course notes

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1 Digital Computation and Quantum Mechanics

1.1 A digital computation

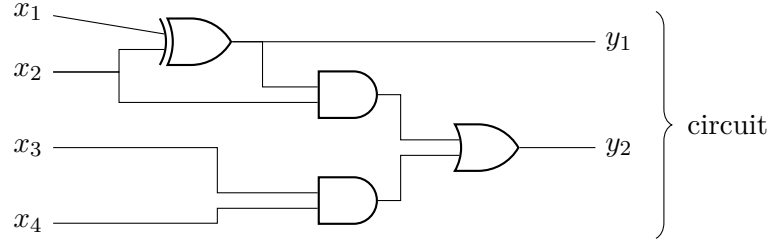


Figure 1.1: A classical digital circuit: inputs, gates and outputs.

The elements of a computation are the **input**, the **gates** and the **output**. An **algorithm** is a sequence of steps that must be done in order to do something (solve a problem), given an instance of a certain size (the initial conditions).

- Physical realisation of the registers (input/output): any two-level system.
- Readout: measurement.
- Loading: initialisation.

1.2 Elements from the quantum-mechanical point of view

register of bit strings	→	state space
manipulation	→	unitary evolution
readout	→	measurement (inner products)
bit → bit string	→	tensor products

Postulates of quantum mechanics.

1. The state of any *closed* physical system is completely described by a vector $|\psi\rangle$ belonging to a complex Hilbert space $\mathcal{H}$.
2. The evolution of a quantum state is given by the application of a *unitary* transformation, $UU^\dagger = U^\dagger U = \mathbb{I}$ (which preserves the scalar product, i.e. lengths):

$$|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle.$$

Equivalently,

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle,$$

Schrödinger's equation, where the Hermitian operator $H = H^\dagger$ is the Hamiltonian of the system.

3. Measurement (readout): physical quantities are represented by Hermitian operators (observables), and the result of a measurement is one of the eigenvalues of the operator. The probability of finding the system in the eigenstate $|\varphi_i\rangle$ of the observable after measuring the state $|\psi\rangle$ is

$$P_i = |\langle \varphi_i | \psi \rangle|^2;$$

after the measurement the state collapses to the eigenstate of the measured value.

4. Composite systems are described by tensor products, $\mathbb{C}^N \otimes \mathbb{C}^M \cong \mathbb{C}^{NM}$.

Spectral theorem. Given a normal matrix M (e.g. unitary or Hermitian) there exists an orthonormal basis $\{|\varphi_i\rangle\}$ such that

$$M = \sum_i \lambda_i |\varphi_i\rangle\langle\varphi_i|,$$

i.e. M can be diagonalised, and the $|\varphi_i\rangle$ are obtained by applying a unitary transformation to another basis.

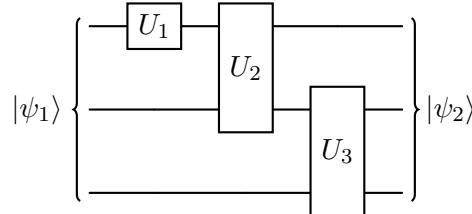
Tensor products. A way of representing them is

$$\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_N \end{pmatrix} \otimes \vec{\beta} = \begin{pmatrix} \alpha_1 \vec{\beta} \\ \vdots \\ \alpha_N \vec{\beta} \end{pmatrix}, \quad \begin{pmatrix} \alpha_{11} & \cdots & \alpha_{1N} \\ \vdots & \ddots & \vdots \\ \alpha_{N1} & \cdots & \alpha_{NN} \end{pmatrix} \otimes B = \begin{pmatrix} \alpha_{11}B & \cdots & \alpha_{1N}B \\ \vdots & \ddots & \vdots \\ \alpha_{N1}B & \cdots & \alpha_{NN}B \end{pmatrix}.$$

1.3 Quantum computing vulgaris

- **Qubit:** $|\psi\rangle \in \mathbb{C}^2$, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $\langle\psi|\psi\rangle = 1$.
- **Register:** $(\mathbb{C}^2)^{\otimes N} = \mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2$ (N times), with computational basis $|00\cdots 0\rangle, |00\cdots 1\rangle, \dots$
- **State:** $|\psi^{(N)}\rangle = \sum_{b_1, \dots, b_N} \alpha_{b_1 \dots b_N} |b_1 b_2 \cdots b_N\rangle$.
- **Evolution of the register:** any unitary map U .
- **Readout:** measurement.
- **Local unitary operation:** for $U \in M_2(\mathbb{C})$ and $\mathbb{K}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, the operation acting on qubit k is $U^{(k)} = \mathbb{K}_2 \otimes \cdots \otimes U \otimes \cdots \otimes \mathbb{K}_2$ (U at site k).

Quantum circuits. Gates in the same time slice combine by tensor products; successive slices combine by operator composition (read from right to left in formulas):



$$|\psi_2\rangle = (\mathbb{K} \otimes U_3)(U_2 \otimes \mathbb{K})(U_1 \otimes \mathbb{K} \otimes \mathbb{K}) |\psi_1\rangle.$$

Some special gates.

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma^x = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^y = Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^z = Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (\text{Pauli matrices}),$$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (\text{Hadamard}),$$

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \quad (\pi/8 \text{ gate}), \quad Z_\phi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \quad (z\text{-rotation}).$$

Two-qubit gates:

$$\begin{aligned}
\begin{array}{c} \text{---} \bullet \text{---} \\ | \\ \text{---} \oplus \text{---} \end{array} &= \begin{pmatrix} \mathbb{I} & 0 \\ 0 & X \end{pmatrix} & \text{CNOT } |b_1 b_2\rangle = |b_1, b_1 \oplus b_2\rangle \\
\begin{array}{c} \text{---} \bullet \text{---} \\ | \\ \text{---} \boxed{U} \text{---} \end{array} &= \begin{pmatrix} \mathbb{I} & 0 \\ 0 & U \end{pmatrix} & CU |b_1\rangle |b_2\rangle = |b_1\rangle U^{b_1} |b_2\rangle \\
\begin{array}{c} \text{---} \times \text{---} \\ | \\ \text{---} \times \text{---} \end{array} &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \text{SWAP } |b_1 b_2\rangle = |b_2 b_1\rangle
\end{aligned}$$

Statement (Universality). The set $G_1 = \{\text{CNOT}\} \cup SU(2)$ (CNOT plus all single-qubit gates) is universal. The set $\{H, T\}$ can approximate any single-qubit rotation.

2 Computability, Complexity and the First Quantum Algorithms

2.1 Recap: gate sets

- Universal: $G = \{\text{CNOT}\} \cup U(2)$ (all single-qubit gates).
- Approximately universal: $\tilde{G} = \{\text{CNOT}, H, T\}$.
- Computationally approximately universal: $G' = \{\text{Toffoli}, H\}$; these gates are real.

2.2 Computability and computational complexity theory

A problem is **computable** if there exists a Turing machine that solves it.

Statement (Church–Turing thesis). Every computable function is computable on a deterministic Turing machine.

Not all problems are computable (e.g. the halting problem, Hilbert’s tenth problem). Can a quantum computer do better? No: every problem solvable on a quantum computer is also solvable on a classical computer. The question is how *fast* each one is.

Complexity measures.

- Gate complexity: number of gates as a function of the input size n .
- Space complexity: number of qubits needed.
- Depth complexity: number of layers (related to parallelisation).

$\mathcal{O}$ denotes an upper bound and Ω a lower bound.

A **BQP** problem is a decision problem (yes/no answer) that can be solved efficiently with bounded error on a quantum computer, measuring a single qubit at the end. Strong beliefs:

$$P \neq NP, \quad BQP \neq BPP, \quad NP \not\subseteq BQP.$$

2.3 Quantum algorithms and their main properties

Deutsch–Jozsa. Given $f : \{0, 1\}^n \rightarrow \{0, 1\}$, promised to be either constant or balanced, decide which. A deterministic classical algorithm needs $2^{n-1} + 1$ queries in the worst case; a quantum computer needs a single query of the oracle

$$U_f |\vec{x}\rangle |b\rangle = |\vec{x}\rangle |b \oplus f(\vec{x})\rangle.$$

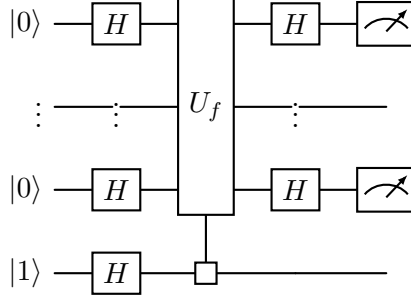
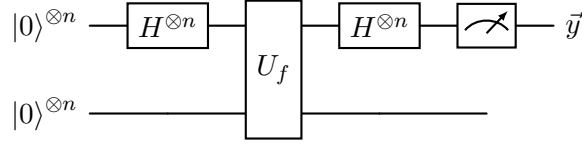


Figure 2.1: Deutsch–Jozsa circuit. f is balanced iff the measured register is not $|\vec{0}\rangle$.

The function is constant if and only if the first register is measured in $|\vec{0}\rangle$.

Simon’s problem. Given $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$ such that there exists $\vec{s} \in \{0, 1\}^n$ with $f(\vec{x}) = f(\vec{y})$ iff $\vec{y} = \vec{x}$ or $\vec{y} = \vec{x} \oplus \vec{s}$, find $\vec{s}$. Classically $\Omega(2^{n/2})$ queries are needed; quantumly $\mathcal{O}(n)$ runs of the circuit

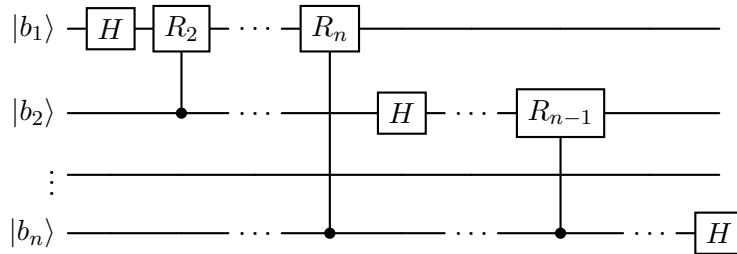


each produce a random $\vec{y}$ with $\vec{s} \cdot \vec{y} = 0 \pmod{2}$, and $n - 1$ independent such equations determine $\vec{s}$.

Quantum Fourier transform. The discrete Fourier transform

$$y_k := \sum_{j=0}^{N-1} x_j e^{-2\pi i j k / N}$$

is given by a matrix M_{kj} which, once normalised by $1/\sqrt{N}$, is unitary. On n qubits it is implemented by the circuit



with

$$R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i / 2^k} \end{pmatrix},$$

followed by a reversal of the order of the qubits.

Quantum period finding. Let $f : \{0, \dots, L-1\} \rightarrow \{0, \dots, L-1\}$ be periodic, $f(x+T) = f(x)$, with $L = 2^k$ and $T < \sqrt{L/2}$.

1. Prepare $|\psi\rangle = \frac{1}{\sqrt{L}} \sum_{x=0}^{L-1} |x\rangle_1 |f(x)\rangle_2$.
2. (Optional) measure register 2, obtaining $y_0 = f(x_0)$:

$$|\psi_0\rangle = \frac{1}{\sqrt{L/T}} \sum_{t=0}^{L/T-1} |x_0 + tT\rangle |f(x_0)\rangle.$$

3. Apply the QFT on register 1: the outcome distribution is sharply peaked at multiples of L/T .
4. Recover T with the continued-fraction algorithm.

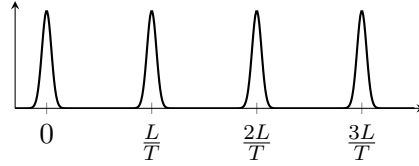


Figure 2.2: Outcome probabilities after the QFT: peaks at multiples of L/T .

Statement (Shor). If one can find the period of $f_{a,N}(x) = a^x \bmod N$ for a randomly chosen a , then one can find the factors of N efficiently.

With $\log N$ the number of digits, the best known classical algorithm (number field sieve) takes $\mathcal{O}(e^{c \sqrt[3]{\log N} (\log \log N)^{2/3}})$ operations, while Shor's algorithm takes $\mathcal{O}((\log N)^2 \log \log N \dots)$.

3 Variational Algorithms

Outline: introduction; the variational paradigm; variational algorithms; trial states; cost functions; optimisation.

3.1 Introduction

- Current devices: 10–100 qubits, noisy, error rates $\sim 10^{-2}$ – 10^{-3} .
- Needed for a speed-up: 100–200 qubits with error rates 10^{-6} – 10^{-9} . To achieve this with quantum error correction one needs $\sim 10^5$ physical qubits.

The current situation is called **NISQ** (Noisy Intermediate-Scale Quantum devices). Variational algorithms are a trade-off between performance and cost.

3.2 The variational paradigm

Given a state $|\psi\rangle$ and a Hermitian operator O with lowest eigenvalue O_0 ,

$$\langle O \rangle = \frac{\langle \psi | O | \psi \rangle}{\langle \psi | \psi \rangle} \geq O_0, \quad (3.1)$$

and if $O |\psi_0\rangle = O_0 |\psi_0\rangle$, then $\langle O \rangle = O_0$ iff $|\psi\rangle = |\psi_0\rangle$. Strategy: prepare many states $\{|\psi_i\rangle\}$, evaluate (3.1) and keep the $|\psi_i\rangle$ that minimises it. In practice $|\psi(\vec{\theta})\rangle$ is parametrised by classical values $\vec{\theta}$ of a unitary $U(\vec{\theta})$:

$$|\psi(\vec{\theta})\rangle = U(\vec{\theta}) |0\rangle,$$

the **ansatz** (trial wave function). A general N -qubit state has $N_{\text{par}} = 2 \cdot 2^N - 2 = \mathcal{O}(2^N)$ real parameters, while variational algorithms use $N_{\text{par}} \sim \text{poly}(N)$ (the general problem is QMA-hard). Open questions: how to construct $U(\vec{\theta})$, and which problems can be solved.

3.3 Variational algorithm

- (i) Initial state (e.g. $|0\rangle^{\otimes N}$ or $|+\rangle^{\otimes N}$).
- (ii) Parametrised quantum circuit $U(\vec{\theta}) = U(\theta_0)U(\theta_1) \cdots U(\theta_n)$.
- (iii) Cost function $f(\vec{\theta})$ and an evaluation method.
- (iv) Classical optimisation.

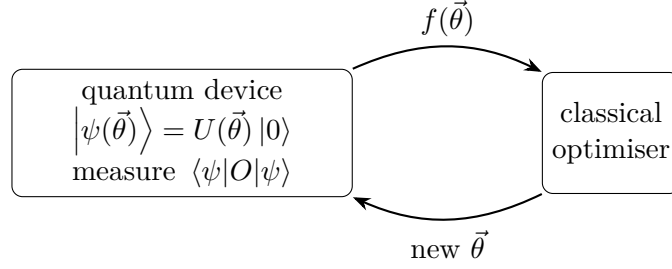


Figure 3.1: Hybrid quantum–classical loop of a variational algorithm.

3.4 Continuously parametrised unitaries

The discrete set $\{H, \text{CNOT}, T, X\}$ can be complemented by continuously parametrised gates generated by Pauli operators $P \in \{X, Y, Z\}$, which satisfy $P^\dagger P = P P^\dagger = \mathbb{I}$ (unitary), $P = P^\dagger$ (Hermitian), $P^2 = \mathbb{I}$:

$$U_P(\theta) = e^{-i\theta P} = \mathbb{I} \cos \theta - iP \sin \theta.$$

For example

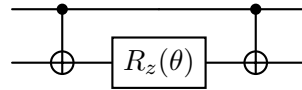
$$U_Z(\theta) = \begin{pmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{pmatrix} = e^{-i\theta} \begin{pmatrix} 1 & 0 \\ 0 & e^{2i\theta} \end{pmatrix} = R_z(\theta), \quad U_X(\theta) = \mathbb{I} \cos \theta - iX \sin \theta = H U_Z(\theta) H,$$

using $HZH = X$, $HXH = Z$. In general $e^{iW^\dagger P W} = W^\dagger e^{iP} W$ for unitary W .

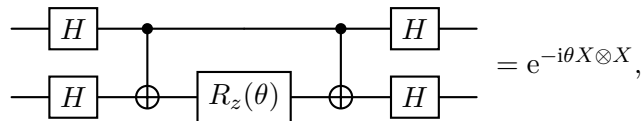
Two-qubit rotations. For $P = ZZ = Z \otimes Z$ (and in general strings like $Z_0 X_3 = Z_0 \otimes \mathbb{I} \otimes \mathbb{I} \otimes X_3$),

$$U_{ZZ}(\theta) = e^{-i\theta Z \otimes Z} = \mathbb{I} \otimes \mathbb{I} \cos \theta - iZ \otimes Z \sin \theta = \text{diag}(e^{-i\theta}, e^{i\theta}, e^{i\theta}, e^{-i\theta}),$$

which is implemented by



Conjugating with Hadamards turns ZZ into XX :



and longer strings such as $e^{-i\theta X_1 X_3}$ are obtained with SWAP gates or longer CNOT ladders.

3.5 Trial states

Starting from a solvable H_0 , we want the ground state of $H_1 = H_0 + \sum_i \lambda_i V_i$.

- (i) **Perturbative ansatz**: $U(\vec{\theta}) = \prod_i e^{i\theta_i V_i}$, $V_i \in \mathcal{P}$ (Pauli strings).
- (ii) **Quantum alternating operator ansatz** (QAOA): $U(\vec{\theta}) = \prod_k e^{i\theta_{2k} H_0} e^{i\theta_{2k+1} H_1}$.
- (iii) **Hardware-efficient ansatz**: layers of single-qubit rotations and entangling gates native to the device.

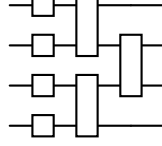


Figure 3.2: Hardware-efficient ansatz (schematic).

3.6 Cost functions and optimisation

(Variational Quantum Eigensolvers.) Pauli strings $\{\mathbb{I}, X, Y, Z\}^{\otimes N}$ form a basis of the (real) vector space of Hermitian matrices. A standard VQE cost function is

$$f_O(\vec{\theta}) = \langle \psi(\vec{\theta}) | O | \psi(\vec{\theta}) \rangle = \sum_{i=1}^{4^N} h_i \langle \psi(\vec{\theta}) | P_i | \psi(\vec{\theta}) \rangle, \quad O = \sum_i h_i P_i.$$

Gradient methods:

- **Parameter-shift rule**: for gates of the form $e^{-i\theta_j P/2}$,

$$\frac{\partial f_O}{\partial \theta_j} = \frac{f_O(\vec{\theta} + \frac{\pi}{2} \vec{e}_j) - f_O(\vec{\theta} - \frac{\pi}{2} \vec{e}_j)}{2}.$$

- **Stochastic techniques** (finite differences along a random direction $\vec{\Delta}$):

$$\frac{\partial f_O}{\partial \theta_j} \simeq \frac{f_O(\vec{\theta} + c\vec{\Delta}) - f_O(\vec{\theta} - c\vec{\Delta})}{2c\Delta_j}.$$

4 Hamiltonian Simulation

Given H , $|\psi(t_0)\rangle$ and $\Delta t = t_1 - t_0$, compute

$$|\psi(t_1)\rangle := e^{-iH\Delta t} |\psi(t_0)\rangle,$$

which follows from the Schrödinger equation $i\partial_t |\psi\rangle = H |\psi\rangle$ ($\hbar = 1$). (There is also the time-independent Schrödinger equation $H |\psi\rangle = E |\psi\rangle$.)

Quantum version.

- Input: an n -qubit register in state $|\psi_0\rangle$ (or $U|0\rangle = |\psi_0\rangle$); a succinct (classical) description of H , or an oracle computing the entries of H (usually sparse); a time t .
- Output: an n -qubit register in state $|\tilde{\psi}_1\rangle$ such that $\| |\psi_1\rangle - |\tilde{\psi}_1\rangle \| < \varepsilon$.

Goal: algorithms with cost $\text{poly}(N)$ or $\text{poly}(\log N)$. This can be achieved with Trotterisation-based algorithms.

4.1 Path to Trotterisation-based algorithms

1. Single-qubit Hamiltonians. Assume the gate set $G = \{H, \text{CNOT}, \text{CZ}, Z_\phi\}$ with $Z_\phi = \text{diag}(1, e^{i\phi})$, and $\hat{H} = \mathbb{K}^{\otimes(k-1)} \otimes Z \otimes \mathbb{K}^{\otimes(n-k)}$. Then

$$e^{-iZt} = \underbrace{e^{-it}}_{\text{global phase}} \begin{pmatrix} 1 & 0 \\ 0 & e^{2it} \end{pmatrix} = e^{-it} Z_{2t},$$

so $e^{-iZ\theta/2} = Z_\theta$ up to a global phase. For a general Pauli $\hat{H} = P \in \{\mathbb{K}, X, Y, Z\}$ we use the

Statement (Lemma). For all $U, W \in U(N)$, $e^{-iUWU^\dagger t} = Ue^{-iWt}U^\dagger$.

Hence $HZ_\phi H^\dagger = X_\phi$, and with $S = \text{diag}(1, i)$,

$$HZH = X, \quad SHZHS^\dagger = Y.$$

For $\hat{H} = \mathbb{K}^{\otimes k} \otimes P \otimes \mathbb{K}^{\otimes(n-k-1)}$ and any U , $e^{-i(\mathbb{K}^{\otimes k} \otimes U \otimes \mathbb{K})t} = \mathbb{K}^{\otimes k} \otimes e^{-iUt} \otimes \mathbb{K}^{\otimes(n-k-1)}$.

2. Two-body terms.

$$e^{-iZ \otimes Z \theta/2} = \text{diag}(e^{-i\theta/2}, e^{i\theta/2}, e^{i\theta/2}, e^{-i\theta/2}) = e^{-i\theta/2} \begin{pmatrix} Z_\theta & 0 \\ 0 & XZ_\theta X \end{pmatrix},$$

implemented by $\text{CNOT}(\mathbb{K} \otimes Z_\theta) \text{CNOT}$ (up to a global phase). Non-adjacent strings are moved together with SWAPs, e.g.

$$(\text{SWAP}_{12} \otimes \mathbb{K})(X \otimes \mathbb{K} \otimes Y)(\text{SWAP}_{12} \otimes \mathbb{K}) = \mathbb{K} \otimes X \otimes Y.$$

Thus CNOT , Z_ϕ and H suffice for the evolution under any Pauli string.

Physically and computationally relevant Hamiltonians. Most are **k -local Hamiltonians**, $\hat{H} = \sum_i \hat{H}_i$, where each $\hat{H}_i$ is a k -local Pauli string. Their evolution is approximated by Trotterisation, $e^{-i\sum_i \hat{H}_i t} \simeq (\prod_i e^{-i\hat{H}_i t/r})^r$.

5 Quantum Phase Estimation

Outline: the QPE problem; phase kickback; single-ancilla QPE for eigenstates; t -ancilla QPE for mixed states; single-ancilla QPE for mixed states.

5.1 The quantum phase estimation problem

“Local Hamiltonian eigenvalue sampling” (LHES).

- Input: $\hat{H} = \sum_i \hat{H}_i$, where the $\hat{H}_i$ are k -local fermionic or qubit operators and the sum has $\text{poly}(N)$ terms (N the system size); $\varepsilon, \delta \in \mathbb{R}$ with $1/\varepsilon, 1/\delta \in \mathcal{O}(\text{poly}(N))$; a known starting state $|\psi_0\rangle = \sum_j a_j |E_j\rangle$, where the $|E_j\rangle$ are eigenstates of $\hat{H}$ with eigenvalues E_j and the amplitudes a_j are not known; a target j .
- Problem: estimate E_j to precision ε with probability $1 - \delta$ in time $\text{poly}(N) |a_0|^{-2}$.

LHES as stated is BQP-complete; without the starting state $|\psi_0\rangle$ (and the $|a_0|^{-2}$ prefactor in the time) it is QMA-hard.

5.2 Phase kickback

- Quantum computer operations correspond to unitaries $\hat{U} \in \mathbb{C}^{2^n} \otimes \mathbb{C}^{2^n}$, $\hat{U}^\dagger \hat{U} = \mathbb{I}$.
- $\hat{H}$ is Hermitian, $\hat{H}^\dagger = \hat{H}$.
- Trick: $\hat{U} = e^{i\hat{H}t}$ is a one-to-one correspondence, and $\hat{U}$ and $\hat{H}$ share eigenvectors: $\hat{H} |E_j\rangle = E_j |E_j\rangle \iff \hat{U} |E_j\rangle = e^{iE_j t} |E_j\rangle$.
- $\hat{U} |\psi_0\rangle = \sum_j a_j e^{iE_j t} |E_j\rangle$.

How do we access the values $e^{iE_j t}$? If U is unitary, so is the controlled operation

$$CU = |0\rangle\langle 0| \otimes \mathbb{I} + |1\rangle\langle 1| \otimes U = \begin{pmatrix} \mathbb{I} & 0 \\ 0 & U \end{pmatrix}.$$

Starting with an ancilla in $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$,

$$\begin{aligned} CU \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \sum_j a_j |E_j\rangle &= \frac{1}{\sqrt{2}} \left(|0\rangle \sum_j a_j |E_j\rangle + |1\rangle U \sum_j a_j |E_j\rangle \right) \\ &= \sum_j \frac{1}{\sqrt{2}} (|0\rangle + e^{iE_j t} |1\rangle) a_j |E_j\rangle =: |\psi_c(t)\rangle : \end{aligned}$$

the phase is “kicked back” onto the ancilla (**phase kickback**). Then

$$\langle \psi_c(t) | (X + iY)_{\text{anc}} \otimes \mathbb{I} | \psi_c(t) \rangle = 2 \langle \psi_c(t) | (|0\rangle\langle 1|) \otimes \mathbb{I} | \psi_c(t) \rangle = \sum_j |a_j|^2 e^{iE_j t}.$$

How can this be implemented on a quantum computer? (a) How does one make CU ? (b) How does one measure $\langle X + iY \rangle$?

(a) Controlled Pauli-string evolutions are obtained from the circuits of the previous section by controlling only the central rotation, e.g. for $e^{i\theta X_1 X_2 Y_3}$ the basis changes and CNOT ladders are left uncontrolled and only the Z rotation is controlled by the ancilla.

5.3 Single-ancilla QPE for eigenstates

(b) $\langle (X + iY) \otimes \mathbb{I} \rangle = \langle X \otimes \mathbb{I} \rangle + i \langle Y \otimes \mathbb{I} \rangle$.

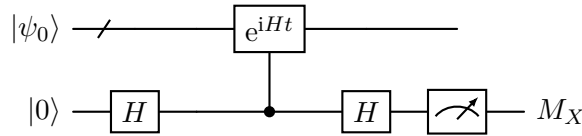


Figure 5.1: Measuring $\langle X \otimes \mathbb{I} \rangle$. For $\langle Y \otimes \mathbb{I} \rangle$ apply $S^\dagger$ before the final Hadamard.

The outcome probabilities are

$$P(M_X = 0) = \frac{1}{2}(1 + \langle X \otimes \mathbb{I} \rangle), \quad P(M_X = 1) = \frac{1}{2}(1 - \langle X \otimes \mathbb{I} \rangle),$$

so $P(M_X = 0) - P(M_X = 1) = \langle X \otimes \mathbb{I} \rangle$, and similarly $P(M_Y = 0) - P(M_Y = 1) = \langle Y \otimes \mathbb{I} \rangle$. Hence

$$\begin{aligned} \langle (X + iY) \otimes \mathbb{I} \rangle &= P(M_X = 0) - P(M_X = 1) + i(P(M_Y = 0) - P(M_Y = 1)) \\ &= \sum_j |a_j|^2 e^{iE_j t} =: g(t), \end{aligned}$$

where k is the number of applications of e^{iHt} .

After M preparations of $|\psi_c(t)\rangle$ and measurements in the X (Y) basis we estimate $P(M_X = 0)$, with variance $V(P(M_X = 0)) = P(M_X = 0)(1 - P(M_X = 0))/M \sim \mathcal{O}(M^{-1})$.

Assume $a_j = \delta_{j,0}$, so $g(k) = e^{iE_0 kt}$. The estimate is

$$\tilde{E}_0 = \frac{1}{kt} \text{Arg } \tilde{g}(k) = \frac{1}{kt} \arctan \frac{\langle Y \otimes \mathbb{K} \rangle}{\langle X \otimes \mathbb{K} \rangle}$$

($\text{Im } g$ over $\text{Re } g$), defined up to multiples of $2\pi/kt$ (how to fix this ambiguity is addressed below). By error propagation, with $Q = \langle Y \otimes \mathbb{K} \rangle / \langle X \otimes \mathbb{K} \rangle$,

$$\text{Var}(E_0) = \left| \frac{\partial E_0}{\partial Q} \right|^2 \text{Var}(Q) = \frac{1}{k^2 t^2 (1 + Q^2)^2} \text{Var}(Q),$$

which gives

$$\text{Var}(E_0) \sim \mathcal{O}((kt)^{-2} M^{-1}), \quad \Delta E_0 \sim \mathcal{O}((kt)^{-1} M^{-1/2}).$$

No fast-forwarding theorem. For an arbitrary Hamiltonian H , the cost of implementing e^{iHkt} scales as $\mathcal{O}(kt)$: one applies the controlled e^{iHt} k times. The total cost (time on the quantum device) is then $T \sim Mkt$. Language:

- $\varepsilon \sim T^{-1/2}$: sampling-noise limited;
- $\varepsilon \sim T^{-1}$: Heisenberg limited.

Robust phase estimation.

1. Estimate $g(k)$ for $k = 2^l$, $l \in \{0, \dots, L\}$, using M_l measurements each. Fix t such that $tE_0 < 2\pi$.
2. Estimate for each l

$$\tilde{F}_l = \frac{1}{2^l t} \arctan \frac{\text{Im } g(2^l)}{\text{Re } g(2^l)} = E_0 + \frac{2\pi q_l}{2^l t}, \quad q_l \in \mathbb{Z},$$

with error $\sim 1/(\sqrt{M_l} 2^l t)$.

3. Set $\tilde{E}_0 = \tilde{F}_0$; at each subsequent l choose q_l to minimise $\left| \tilde{F}_l + \frac{2\pi q_l}{2^l t} - \tilde{E}_{l-1} \right|$ and fix $\tilde{E}_l = \tilde{F}_l + \frac{2\pi q_l}{2^l t}$.

6 The Quantum Fourier Transform and Multi-Ancilla QPE

6.1 The quantum Fourier transform

$$U_{\text{QFT}} |n\rangle = \frac{1}{2^{l/2}} \sum_{m=0}^{2^l-1} e^{2\pi i mn/2^l} |m\rangle$$

on l qubits. It is achieved by the circuit of Section 2.3 with the controlled phase gates R_k defined there, using $\mathcal{O}(l^2)$ gates, compared with $\mathcal{O}(l 2^l)$ operations for the classical fast Fourier transform: an exponential quantum speed-up (of the transform itself).

6.2 Multi-ancilla QPE

Instead of processing the QPE data on a classical computer, one can apply a QFT to a register of l ancillas. After Hadamards on the ancillas and controlled evolutions e^{iHt2^r} controlled by ancilla r ,

$$|\Lambda\rangle = \frac{1}{2^{l/2}} \sum_n e^{iE_j n t} |n\rangle \sum_j a_j |E_j\rangle \longrightarrow U_{\text{QFT}}^\dagger |\Lambda\rangle = \frac{1}{2^l} \sum_{j,m,n} a_j e^{-in(\frac{2\pi m}{2^l} - E_j t)} |m\rangle |E_j\rangle.$$

The sum over n is peaked where $\frac{2\pi m}{2^l} = E_j t$, so

$$U_{\text{QFT}}^\dagger |\Lambda\rangle \approx \sum_j a_j |\tilde{E}_j\rangle |E_j\rangle, \quad \tilde{E}_j = \frac{E_j t 2^l}{2\pi},$$

where $|\tilde{E}_j\rangle$ is written in the computational basis: measuring the ancilla register gives E_j with probability $|a_j|^2$.

7 Strongly Correlated Systems

Definition. A system of many particles where particle–particle interactions are strong enough to require consideration in any description (for accuracy and error estimates). Example: electrons in a molecule, where the interaction can be neither ignored nor treated perturbatively. A quantum computer is itself a strongly correlated system. They are relevant in chemistry, biochemistry, materials science, condensed matter, nuclear physics, etc. (many-body physics).

The wave function is $|\psi\rangle := \psi(\vec{r}_\alpha)$, $\alpha \in \mathcal{B}$, and an interacting Hamiltonian includes interactions between particles. Infinite systems cannot be modelled by any computer: the solution is to project the Hamiltonian onto a space of smaller dimension.

8 Strongly Correlated Systems II

Outline: symmetries; identical particles (fermions and bosons, Slater determinants, creation/annihilation operators, Majorana operators).

8.1 Symmetries

A system is symmetric with respect to some quantity s if its associated operator S commutes with the Hamiltonian:

$$s \text{ is a symmetry} \iff [S, H] = 0.$$

Therefore, if $|E_j\rangle$ is an eigenstate of H , so is $S|E_j\rangle$: $H(S|E_j\rangle) = SH|E_j\rangle = E_j(S|E_j\rangle)$. Two options:

- $S|E_j\rangle = e^{i\phi} |E_j\rangle$ (non-degenerate level);
- $S|E_j\rangle = \sum_{j'} a_{j'} |E_{j'}\rangle$ with $E_{j'} = E_j$ whenever $a_{j'} \neq 0$.

One can find a representation of the eigenstates $|E_j\rangle \rightarrow |\tilde{E}_j\rangle$ such that the $|\tilde{E}_j\rangle$ are eigenstates of both S and H (it may not be local).

8.2 Identical particles

Systems of identical particles are symmetric under the interchange of two particles. With $|\psi_{\alpha_1, \dots, \alpha_N}\rangle := \psi(\vec{r}_1, \dots, \vec{r}_N)$,

$$S_p^{ij} \psi(\vec{r}_1, \dots, \vec{r}_i, \dots, \vec{r}_j, \dots, \vec{r}_N) := \psi(\vec{r}_1, \dots, \vec{r}_j, \dots, \vec{r}_i, \dots, \vec{r}_N).$$

Since $(S_p^{ij})^2 = \mathbb{I}$, $S_p^{ij} \psi = \pm \psi$: + for bosons, – for fermions. The many-body wave function can be written as

$$|\psi_{\alpha_1 \dots \alpha_N}\rangle = \sum_{\alpha_1, \dots, \alpha_N} a_{\alpha_1 \dots \alpha_N} \psi_{\alpha_1}(\vec{r}_1) \cdots \psi_{\alpha_N}(\vec{r}_N),$$

where the symmetry is encoded in the coefficients, $S_p^{ij} a_{\alpha_1 \dots \alpha_i \dots \alpha_j \dots \alpha_N} = \pm a_{\alpha_1 \dots \alpha_j \dots \alpha_i \dots \alpha_N}$.

Slater determinants. An antisymmetric wave function can be written as

$$|\psi_{\alpha_1 \dots \alpha_N}^{\text{as}}\rangle = \frac{1}{\sqrt{N!}} \det \begin{pmatrix} \psi_{\alpha_1}(\vec{r}_1) & \cdots & \psi_{\alpha_1}(\vec{r}_N) \\ \vdots & \ddots & \vdots \\ \psi_{\alpha_N}(\vec{r}_1) & \cdots & \psi_{\alpha_N}(\vec{r}_N) \end{pmatrix},$$

which can be computed in polynomial time.

Eigenstates of S_p^{ij} are not necessarily eigenstates of H . If H does not contain particle–particle interactions, the ground state can be written as a single Slater determinant.

From here on $N \rightarrow \eta$ for the number of particles. With η particles and N single-particle basis functions there are $\binom{N}{\eta}$ many-body states, which form an orthonormal basis for the set of all antisymmetric wave functions. What do operators look like on this space, and what is a “nice” representation of the Slater determinants?

8.3 Second quantisation

Occupation-number states $|\vec{n}\rangle := |n_1\rangle |n_2\rangle \cdots |n_N\rangle$. It is natural to define annihilation operators

$$c_i |n_1, \dots, n_i, \dots, n_N\rangle = (-1)^{\sum_{j < i} n_j} n_i |n_1, \dots, n_i - 1, \dots, n_N\rangle,$$

where the sign accounts for the antisymmetry, and creation operators $c_i^\dagger$. They satisfy

$$\begin{aligned} \text{fermions: } \quad & \{c_i, c_k\} = c_i c_k + c_k c_i = 0, & \{c_i, c_k^\dagger\} &= \delta_{ik}, \\ \text{bosons: } \quad & [c_i, c_k] = c_i c_k - c_k c_i = 0, & [c_i, c_k^\dagger] &= \delta_{ik}. \end{aligned}$$

$n_i := c_i^\dagger c_i$ are the **number operators**. The projectors of the first-quantised Hamiltonian become one-body terms $c_i^\dagger c_j$ (a particle hopping from j to i) and two-body terms $c_i^\dagger c_j^\dagger c_k c_l$:

$$H = \sum_{i,j} h_{ij} c_i^\dagger c_j + \sum_{i,j,k,l} h_{ijkl} c_i^\dagger c_j^\dagger c_k c_l.$$

Hence the expectation values of one- and two-particle operators are contained in the fermionic two-particle **reduced density matrix** (RDM)

$$D_{j_1 j_2}^{i_1 i_2} = \langle \psi | c_{i_1}^\dagger c_{i_2}^\dagger c_{j_1} c_{j_2} | \psi \rangle,$$

and the expectation values of one-particle operators in the 1-RDM, $D_j^i = \langle \psi | c_i^\dagger c_j | \psi \rangle$. (The total number of particles is usually conserved at room temperature.)

A single orbital. For $i = j$, $S_p^{ii}\psi(\vec{r}_i) = -\psi(\vec{r}_i) = \psi(\vec{r}_i)$, so $\psi(\vec{r}_i) = 0$ (Pauli exclusion). There are two basis states, $|0\rangle$ and $|1\rangle$, and

$$c_i = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad c_i^\dagger = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad c_i + c_i^\dagger = \sigma_x, \quad i(c_i^\dagger - c_i) = \sigma_y.$$

However, Pauli matrices on different sites commute, so they do not respect the fermionic anticommutation relations.

Majorana operators (the fermionic equivalent of the Pauli operators):

$$\gamma_{2i} := c_i + c_i^\dagger, \quad \gamma_{2i+1} := i(c_i^\dagger - c_i), \quad \gamma_i^\dagger = \gamma_i, \quad \gamma_i^2 = \mathbb{I}, \quad \{\gamma_i, \gamma_j\} = 2\delta_{ij}.$$

One can map them to Pauli operators, always keeping track of the sign; such a transformation must be non-local.

Jordan–Wigner transformation. Map $\gamma_{2i} \rightarrow X_i$ and $\gamma_{2i+1} \rightarrow Y_i$, but leave a string of Z operators on all qubits to the left of i (using $\{Z_i, X_i\} = \{Z_i, Y_i\} = 0$):

$$\gamma_{2i} = \bigotimes_{j<i} Z_j \otimes X_i, \quad \gamma_{2i+1} = \bigotimes_{j<i} Z_j \otimes Y_i, \quad c_i^\dagger = \frac{1}{2}(\gamma_{2i} - i\gamma_{2i+1}) = \frac{1}{2} \bigotimes_{j<i} Z_j \otimes (X_i - iY_i).$$

9 Strongly Correlated Systems III

References: arXiv:1808.10402, arXiv:1812.09976.

Outline: fermion-to-qubit transforms (Jordan–Wigner, Bravyi–Kitaev, Jiang–Kalev–Mruczkiewicz–Neven (JKMN)); model systems (lattices, Ising model, Heisenberg model, Hubbard model, electronic structure problem); targets for simulation.

9.1 Fermion-to-qubit transformations

Recap: fermionic systems are defined by creation and annihilation operators $c_i^\dagger, c_i$ such that

$$\{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0, \quad \{c_i, c_j^\dagger\} = \delta_{ij},$$

where i is an index/site/orbital. Each orbital corresponds to a single-particle wave function that may be occupied by 0 or 1 particle. We need to transform them to Pauli operators $\{X, Y, Z, \mathbb{I}\}^{\otimes N}$ while maintaining the (anti)commutation relations; it is easiest to work with Majorana operators.

Jordan–Wigner. Order the sites on a line; the Majorana γ_{2i} becomes X_i with a **Jordan–Wigner string** $Z_0 \cdots Z_{i-1}$:

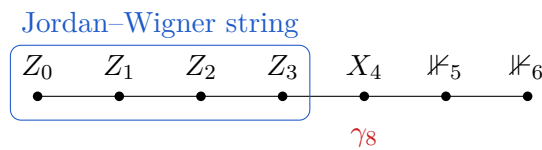


Figure 9.1: Jordan–Wigner image of the Majorana operator $\gamma_8 = \gamma_{2 \cdot 4}$.

Does this maintain the anticommutation rules? For two tensor products of Paulis $P = p_1 \otimes p_2 \otimes \cdots \otimes p_n$ and $Q = q_1 \otimes \cdots \otimes q_n$, $p_i, q_i \in \{X, Y, Z, \mathbb{I}\}$, P and Q commute (anticommute) if an even (odd) number of pairs (p_i, q_i) anticommute. The images above satisfy the Majorana algebra. Drawback: the average local fermionic operator becomes an $\mathcal{O}(N/2)$ -local qubit operator.

Bravyi–Kitaev. Can one reduce the locality of the JW transformation? Find a better graph on which to create the strings: a tree.

- A **graph** is a collection of vertices $\{v_i\}$ and edges $\{e_i\}$.
- A **tree** is a graph in which there is only one path along edges to each vertex.

In a binary tree of depth d each vertex corresponds to a site ($2^d - 1$ sites), and paths to the root have length $\log N$.

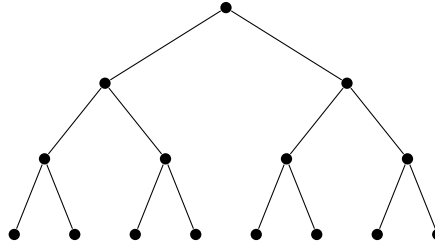


Figure 9.2: Binary tree: each vertex is a site; the path from any vertex to the root has length $\mathcal{O}(\log N)$.

Three rules (which work for any tree) give the qubit image of a Majorana operator: string X 's up the tree to the root; put Z 's on the children of the vertex for γ_{2i+1} ; put Z 's on the children of the parent nodes to the right of the path. Then all Majorana images anticommute as required, and the strings have length $\mathcal{O}(\log N)$. Note: the binary tree is suboptimal; the Fenwick tree is the usual choice.

JKMN transform. Replace the binary tree by a ternary tree. Vertices no longer correspond to sites: vertices with children correspond to qubits and leaves correspond to Majoranas. Label the three edges below each qubit vertex X, Y, Z ; to generate the Pauli string of a Majorana, go from the root to the corresponding leaf and pick up the Pauli operators along the way.

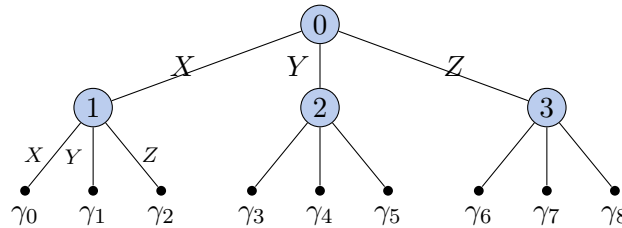


Figure 9.3: Ternary tree with 4 qubits and 9 Majorana leaves. For example $\gamma_1 \mapsto X_0 Y_1$.

9.2 Model systems

Lattice systems. A d -dimensional lattice is an infinite graph in d dimensions with translational symmetry. One often cuts the lattice beyond some edge to obtain a finite system. With clever tricks, geometrically local fermions on a lattice can be mapped to geometrically local qubits (e.g. the Bravyi–Kitaev superfast encoding, BKSF).

The Ising model. On a lattice (or any graph) we write $\langle i, j \rangle$ for vertices sharing an edge (nearest neighbours):

$$H_I = \sum_{\langle i, j \rangle} J_{ij} Z_i Z_j, \quad J_{ij} \in \mathbb{R}.$$

- For an arbitrary choice of J_{ij} , finding the ground-state energy is NP-hard.
- $J_{ij} = -J < 0$: ferromagnet; $J_{ij} = J > 0$: antiferromagnet.
- Optimisation problems can easily be mapped to the Ising model.
- A transverse field $\sum_i h_i X_i$ can be added to make the system quantum.

The Heisenberg model (quantum version of the Ising model):

$$H_H = \sum_{\langle i, j \rangle} J_{ij} (X_i X_j + Y_i Y_j + Z_i Z_j).$$

The number of terms scales as $\mathcal{O}(N)$; for a square lattice each site has 4 edges, i.e. 12 terms each shared with another site, so $\# \text{terms} \simeq 6N$. An on-site potential is often added.

The Hubbard model (spinful fermion model):

$$H = \sum_{\langle i, j \rangle} \sum_{\sigma=\pm} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + \sum_i V_i c_{i+}^\dagger c_{i+} c_{i-}^\dagger c_{i-},$$

with a spin-conserving hopping term and an on-site repulsion (if $V_i > 0$). Each site holds two spin orbitals, and the number of terms is $\mathcal{O}(N)$. It is thought to be a good model for high-temperature superconductivity.

Via Jordan–Wigner (ordering the sites in a snake through the lattice, with two qubits per site labelled (i, j, σ)), the on-site term is

$$c_{ij+}^\dagger c_{ij+} c_{ij-}^\dagger c_{ij-} = n_{ij+} n_{ij-} = \frac{(\mathbb{I} - Z_{ij+})(\mathbb{I} - Z_{ij-})}{4} :$$

no JW strings remain, since they cancel ($Z^2 = \mathbb{I}$). Hopping terms between sites adjacent in the ordering, e.g. $c_{31-}^\dagger c_{21-}$, keep a short string,

$$c_{31-}^\dagger c_{21-} = \frac{1}{4} (X_{31-} - iY_{31-}) Z_{31+} \cdots (X_{21-} + iY_{21-}),$$

giving 2- to 4-local qubit terms. Vertical connections need JW strings between sites along the snake: the maximum length is $\sim \sqrt{N}/2 \cdot 2 \cdot 2 = \sqrt{8N}$ and the average $\sim \sqrt{2N}$.

10 Strongly Correlated Systems IV

Outline: the electronic structure problem; system summary; properties to calculate; the NISQ process diagram; unitary coupled cluster.

10.1 The electronic structure problem

- Not a lattice model: a model of electrons around nuclei in molecules.
- Typically (not always) nuclei are heavy enough to be treated classically (Born–Oppenheimer).

Begin from the Hamiltonian with Coulomb interactions,

$$H = \sum_i \frac{-\hbar^2 \nabla_i^2}{2m_i} + \sum_{i < j} \frac{Z_i Z_j e^2}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_j|},$$

a fermionic system where the sums currently run over nuclei and electrons. Treating the nuclei as classical and static,

$$H = H_0 + \sum_{i \in e^-} \frac{-\hbar^2 \nabla_i^2}{2m} - \sum_{\substack{i \in e^- \\ j \in n}} \frac{Z_j e^2}{4\pi\epsilon_0 |\vec{r}_i - \vec{R}_j|} + \underbrace{\sum_{\substack{i < j \\ \in e^-}} \frac{e^2}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_j|}}_{\text{interacting term}}.$$

Go to a second-quantised basis: take a set of single-particle wave functions $\psi_i(\vec{r})$, form Slater determinants, and associate to each ψ_i a pair $c_i, c_i^\dagger$:

$$H = H_0 + \sum_{i,j} t_{ij} c_i^\dagger c_j + \sum_{i,j,k,l} V_{ijkl} c_i^\dagger c_j^\dagger c_k c_l \quad (\mathcal{O}(N^4) \text{ terms}),$$

where H_0, t_{ij}, V_{ijkl} are numbers:

$$t_{ij} = \int d^3r \psi_i^*(\vec{r}) \left[-\frac{\hbar^2 \nabla^2}{2m} - \sum_k \frac{Z_k e^2}{4\pi\epsilon_0 |\vec{r} - \vec{R}_k|} \right] \psi_j(\vec{r}),$$

$$V_{ijkl} = \int d^3r_1 d^3r_2 \psi_i^*(\vec{r}_1) \psi_j^*(\vec{r}_2) \frac{e^2}{4\pi\epsilon_0 |\vec{r}_1 - \vec{r}_2|} \psi_k(\vec{r}_2) \psi_l(\vec{r}_1).$$

Both are classically tractable to calculate (the cost depends on the choice of basis set). State of the art (2020): instances of up to 12–14 qubits attempted on quantum devices.

10.2 Properties to calculate

What do we want from quantum computations of strongly correlated systems? To reproduce and predict experiments. Quantum mechanics says that all information about a system is kept in expectation values of observables:

- energies of Hamiltonians, in particular ground states;
- expectation values of local observables, e.g. $\sum_i \langle Z_i \rangle$ (total magnetisation) or $\langle n_i \rangle = \langle c_i^\dagger c_i \rangle$ (electronic structure problem);
- derivatives of the above, $\partial \langle O \rangle / \partial \lambda$, $\partial E / \partial \lambda$, as we change external parameters.

Derivatives of expectation values can often be mapped to higher-order derivatives of the energy: with $H \rightarrow H + v_i n_i$,

$$\langle n_i \rangle = \frac{\partial E}{\partial v_i} \Big|_{v_i \rightarrow 0}, \quad \frac{\partial \langle n_i \rangle}{\partial \lambda} = \frac{\partial^2 E}{\partial v_i \partial \lambda} \Big|_{v_i \rightarrow 0}$$

(in the ground state). First-order energy derivatives are obtained with the **Hellmann–Feynman theorem**,

$$\frac{\partial E}{\partial \lambda} = \langle \psi | \frac{\partial H}{\partial \lambda} | \psi \rangle \quad \text{if } |\psi\rangle \text{ is an eigenstate of } H,$$

which does not hold for higher derivatives. Dynamical quantities require measuring $\langle \psi | P_1 e^{iHt} P_2 | \psi \rangle$ or $\langle \psi | e^{-iHt} P_1 e^{iHt} P_2 | \psi \rangle$.

10.3 VQE recap

We need:

- (quantum) a parametrised quantum circuit $U(\vec{\theta})$, the variational ansatz;
- (quantum) a starting state $|\vec{0}\rangle$, not necessarily $|0\rangle$ in the computational basis;
- (classical) a cost function $E(\vec{\theta})$;
- (classical) an optimisation routine varying $\vec{\theta}$ to minimise $E(\vec{\theta})$;
- (quantum) a protocol to estimate $E(\vec{\theta})$ (and possibly its derivatives) on the quantum hardware.

10.4 Unitary coupled cluster

UCC is a VQE for the electronic structure problem (variants exist for spin models, and it extends trivially to other fermionic systems). It is motivated by perturbation theory. Starting from

$$H = \sum_{i,j} t_{ij} c_i^\dagger c_j + \sum_{i,j,k,l} V_{ijkl} c_i^\dagger c_j^\dagger c_k c_l,$$

set $V_{ijkl} = 0$ for now (or approximate the two-body term by a mean field, $c_i^\dagger c_j^\dagger c_k c_l \approx \frac{1}{2} [\langle c_i^\dagger c_l \rangle c_j^\dagger c_k + \langle c_j^\dagger c_k \rangle c_i^\dagger c_l]$). Diagonalising the quadratic Hamiltonian with $c_i = \sum_j u_{ij} \tilde{c}_j$,

$$H = \sum_{i,j} t_{ij} c_i^\dagger c_j = \sum_{\alpha} \varepsilon_{\alpha} \tilde{c}_{\alpha}^\dagger \tilde{c}_{\alpha}$$

(the Hartree–Fock, tight-binding or non-interacting solution). The ground state with N particles is

$$|\psi_{\text{HF}}\rangle = \prod_{\alpha=0}^{N-1} \tilde{c}_{\alpha}^\dagger |\text{vac}\rangle, \quad E_{\text{HF}} = \sum_{\alpha=0}^{N-1} \varepsilon_{\alpha}.$$

Now turn the interactions on: $H = \sum_{\alpha} \varepsilon_{\alpha} c_{\alpha}^\dagger c_{\alpha} + \sum_{\alpha\beta\gamma\delta} \tilde{V}_{\alpha\beta\gamma\delta} c_{\alpha}^\dagger c_{\beta}^\dagger c_{\gamma} c_{\delta}$. Physical intuition: if $\tilde{V} \ll \varepsilon$, the ground state “looks like”

$$|\psi_{\text{HF}}\rangle + \sum_{\substack{\alpha,\beta < N \\ \gamma,\delta \geq N}} \frac{\tilde{V}_{\alpha\beta\gamma\delta}}{\varepsilon_{\alpha} + \varepsilon_{\beta} - \varepsilon_{\gamma} - \varepsilon_{\delta}} c_{\gamma}^\dagger c_{\delta}^\dagger c_{\beta} c_{\alpha} |\psi_{\text{HF}}\rangle + \mathcal{O}\left(\frac{N^2}{(\varepsilon - \varepsilon')^2}\right),$$

which moves particles from filled states ($< N$) to empty states ($\geq N$). This is generalised by the **cluster operator**

$$T = T^{(1)} + T^{(2)} + T^{(3)} + \dots, \quad T^{(1)} = \sum_{\substack{\alpha < N \\ \beta \geq N}} \theta_{\alpha}^{\beta} c_{\beta}^\dagger c_{\alpha}, \quad T^{(2)} = \sum_{\substack{\alpha,\beta < N \\ \gamma,\delta \geq N}} \theta_{\alpha\beta}^{\gamma\delta} c_{\gamma}^\dagger c_{\delta}^\dagger c_{\alpha} c_{\beta}, \quad \dots,$$

with $\mathcal{O}(N^4)$ terms in $T^{(2)}$. What are the θ ? We don’t know; optimise them variationally. Going further, the **coupled cluster** approximation is

$$|\psi(\vec{\theta})\rangle = e^{T(\vec{\theta})} |\psi_{\text{HF}}\rangle,$$

which can be evaluated classically at low truncation, normally to singles and doubles (CCSD): $T_{\text{CCSD}} = T^{(1)} + T^{(2)}$. The lack of de-excitations is necessary for classical convergence, but not physical; we would prefer the unitary version

$$|\psi(\vec{\theta})\rangle = e^{T_{\text{CCSD}}(\vec{\theta}) - T_{\text{CCSD}}^\dagger(\vec{\theta})} |\psi_{\text{HF}}\rangle =: U_{\text{UCCSD}}(\vec{\theta}) |\psi_{\text{HF}}\rangle.$$

For any operator O , $O - O^\dagger$ is anti-Hermitian, so $e^{O - O^\dagger}$ is unitary. (One could also choose $e^{i(T + T^\dagger)}$, but then U is imaginary, giving worse overlap with the states of a real H .)

Complexity. Typically at most one gate per parameter is needed, so $\mathcal{O}(N^4)$ gates is a lower bound. Expanding via Trotterisation,

$$U_{\text{UCCSD}}(\vec{\theta}) \simeq \prod_{\substack{\alpha, \beta \text{ filled} \\ \gamma, \delta \text{ empty}}} \exp \left[\theta_{\alpha\beta}^{\gamma\delta} (c_{\gamma}^{\dagger} c_{\delta}^{\dagger} c_{\alpha} c_{\beta} - c_{\beta}^{\dagger} c_{\alpha}^{\dagger} c_{\delta} c_{\gamma}) \right],$$

and with a Jordan–Wigner transform there are $\mathcal{O}(N^4)$ terms of $\mathcal{O}(N)$ gates each, i.e. $\mathcal{O}(N^5)$ gates. Luckily, one knows how to avoid JW strings during parallelisation, giving $\mathcal{O}(N^3)$ circuit depth; one can also cut terms to make the depth smaller (at a cost in accuracy).

11 Quantum-Enhanced Machine Learning (with Near-Term Devices)

Contents: background; QC meets ML; ML and parametrised circuits; quantum ML with quantum feature spaces.

11.1 Machine learning

Machine learning and AI: where are the boundaries? There are three main (canonical) modes of ML:

- **Supervised learning:** e.g. given labelled images of cats and ducks, classify an image never seen before.
- **Unsupervised learning:** learn the underlying features.
- **Reinforcement learning:** learn correct behaviour (policies) by trial and error.

Supervised learning, the *how*: **regularisation**, i.e. controlling “model complexity” to ensure good generalisation. Machine learning is all about **generalisation performance**, that is, performance beyond the training set.

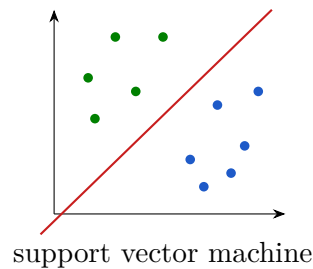


Figure 11.1: A support vector machine separates the classes with a hyperplane. Neural networks instead learn a parametrised function.

When discussing quantum ML, keep an eye on the *what*, the *how* and the *why*. “Objective is subjective.”

11.2 QC meets ML: the big picture

Quantum computing could address three bottlenecks: optimisation, high dimensionality, and hard models.